



AI-Powered Metadata Editor

Schema-agnostic generation, migration and harmonisation of metadata using Large Language Models

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Motivation

Lengthy and tedious task

Metadata creation is costly, inconsistent, and largely manual - a known bottleneck in SDIs and data spaces

Schema heterogeneity

Several standards: ISO 19115, INSPIRE, DCAT, GeoDCAT-AP

Content inconsistency

within the same schema across operators and time

- ✓ This AI-powered metadata editor addresses these challenges within a unified, schema agnostic pipeline.
- ✓ Metadata are automatically extracted from dataset and companion documentation – with a human-in-the-loop validation

Source code published at <https://code.europa.eu/jrc-dsi-lab/geospatial-metadata> under EUPL v.1.2

Three types of metadata attributes

Type A Descriptive

Free-text fields suitable for LLM generation

Examples: title, description, keywords, provenance, temporal coverage

Generated via RAG: semantically oriented queries issued against a vector corpus; results ranked by relevance; context passage assembled to ground the LLM response

Type B Technical

Data-intrinsic properties extracted directly from the dataset

Examples: bounding box, CRS, spatial resolution, file format, geometry type

Extracted via GDAL/OGR from the dataset file, OGC service URL, or gdalinfo/ogrinfo textual output

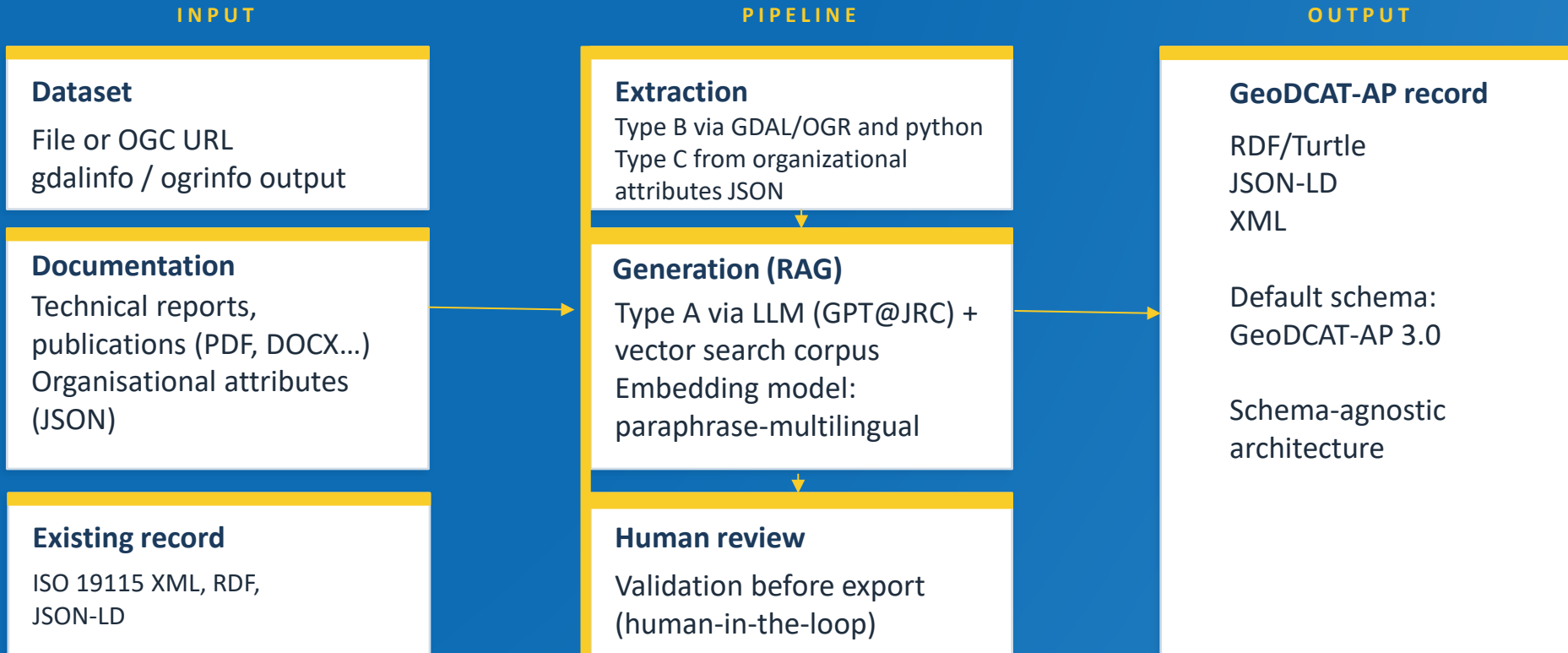
Type C Institutional

Properties defined by the data provider organisation

Examples: publisher, contact point, license, access rights, distribution

Declared by the user via an organizational attributes JSON file

Tool overview





AI-generated content may be inaccurate. Always review and verify metadata before publishing.

Metadata Schema Selection

Choose the metadata standard for your dataset

Schema: **GeoDCAT-AP 3.0.0 - 39 attributes**

Geographic extension of DCAT-AP for spatial datasets

[View Specification](#)

Settings

Configure API credentials and customize prompts

API Credentials

Upload your OpenAI-compatible API credentials to enable AI features.

Status: Not configured

TEMPLATE

UPLOAD

CLEAR

Custom Prompts

Download and customize AI prompts for metadata generation.

Status: Using defaults

DOWNLOAD

UPLOAD

RESET

Type C Parameters (Policy/Organization)

Upload organizational metadata: publisher, contact info, licensing, etc. These values override extracted/generated data. Supports CSV and JSON formats.

Status: Not configured

TEMPLATE

UPLOAD

CLEAR

Add Geospatial Dataset

Add your dataset via file upload or URL

Upload Local File

Supports geospatial (GeoPackage, Shapefile, GeoTIFF, etc.) and tabular (CSV, Excel) formats. For Shapefiles, select all components (.shp, .shx, .dbf, .prj).

UPLOAD DATASET

Load from URL

Paste any URL - the system will auto-detect the type (file, directory, or OGC service)

<https://example.com/data/> or <https://example.com/wfs>

API credentials
e.g. GPT@JRC

1. Download template
2. Fill it in
3. Upload

Organisational
attributes (JSON)

1. Download template
2. Fill it in
3. Upload

You can use
pre-defined
prompts or
customise
your own:

1. Download template
2. Fill it in
3. Upload

Dataset upload, or load from URL



Add Geospatial Dataset

Add your dataset via file upload or URL

Upload Local File

Supports geospatial (GeoPackage, Shapefile, GeoTIFF, etc.) and tabular (CSV, Excel) formats. For Shapefiles, select all components (.shp, .shx, .dbf, .prj).

UPLOAD DATASET

Load from URL

Paste any URL - the system will auto-detect the type (file, directory, or OGC service)

LOAD FROM URL

Supported: Direct files (.tif, .gpkg, .zip), directories, WFS, WCS, OGC API Features

Upload supporting documents describing the datasets: technical reports, publications, scientific papers, existing metadata records



Upload Documents

Upload supporting documents for metadata extraction

Supported formats: PDF, Word, Text, Markdown, HTML, XML, JSON, JSON-LD, RDF, CSV (Turtle, N3, N-Triples, RDF/XML)

Supporting Documents

Dataset documentation from which AI extracts metadata automatically.

- Technical reports & publications
- Existing metadata files (XML, RDF)
- User guides & documentation
- Any PDFs/docs describing the dataset

UPLOAD DOCUMENTS

Hit this button



Uploaded Files (1)

 Abies-alba_mhs.tif ✕

✓ Dataset uploaded: Abies-alba_mhs.tif

EXTRACT METADATA FIELDS

- ✓ Human in the loop validation.
- ✓ User can edit fields manually.
- ✓ Fields have a topological dependency. Once a field is manually edited, dependent fields are flagged as outdated, and AI generation can be triggered manually

Extracted Metadata Fields

Schema: **GeoDCAT-AP 3.0.0** - Geographic extension of DCAT-AP for spatial datasets

33

Total Fields

11

Type A (AI)

10

Type B (Auto)

12

Type C (Policy)

16

Optional Hidden

16 optional fields were hidden because no values were found. Scroll down to "Add Optional Fields" to manually add missing metadata (e.g., Type B fields like resolution or extent).

VALIDATE

EXPORT TURTLE

EXPORT RDF/XML

EXPORT JSON-LD

Access Rights Optional **TYPE C**

Information about who can access the Dataset or an indication of its security status.

Extracted Value From policy

no-limitations

▲ 'no-limitations' not found in Access Rights Vocabularies (EU + INSPIRE) vocabulary

Attribute Schema Optional **TYPE B**

1 source

Description of the attribute schema (fields and types).

Extracted Value Auto-extracted

Raster with 1 bands

Bounding Box Optional **TYPE B**

1 source

Geographic bounding box of the Dataset (minX, minY, maxX, maxY).

Extracted Value Auto-extracted

2600000.0000000005,1400000.0000000023,6700000.0,5450000.000000002

Conforms To Optional **TYPE C**

An established standard to which the Dataset conforms.

Extracted Value From policy

GeoDCAT-AP 3.0.0 <<https://semiceu.github.io/GeoDCAT-AP/releases/3.0.0/>>

Contact Point Mandatory TYPE C

Contact information for enquiries about the Dataset.

Extracted Value From policy

JRC Data Support, jrc-data-support@ec.europa.eu

Coordinate Reference System Optional TYPE B

 1 source

x

Coordinate Reference System used in the Dataset.

Extracted Value Auto-extracted

http://www.opengis.net/def/crs/EPSG/0/3035

Creation Date Optional TYPE B

 1 source

x

The date on which the Dataset was created.

Extracted Value Auto-extracted

2016-08-01

Creator Optional TYPE C

x

The entity primarily responsible for producing the Dataset.

Extracted Value From policy

Joint Research Centre of the European Commission <https://joint-research-centre.ec.europa.eu/index_en>

Description Mandatory TYPE A Generate last

A free-text account of the Dataset.

Extracted Value AI-generated

This dataset provides Abies Alba maximum habitat suitability data. It depicts tree species habitat suitability for Europe in 2006, with a resolution of 1 kilometer. The data have been produced using [Not specified] and sourced from the Joint Research Centre. The data are available in GeoTIFF format and have cell values indicating habitat suitability scores. Potential applications include biodiversity assessment and forest management planning. The dataset can be accessed at https://w3id.org/mtv/FISE-Comm/v01/e01493b/map-MHS, with the following access rights: no-limitations. For enquiries, contact JRC Data Support (jrc-data-support@ec.europa.eu). The dataset is updated with an unknown frequency.

Three operational modes

*The mode is not selected by the user - it is **determined by what is uploaded**.
The processing pipeline is identical in all three cases.*

Creation	INPUT Dataset + Supporting documents + Organisational attributes	HOW Type B extracted automatically. Type A generated via LLM+RAG. A shared prompt template encodes curatorial conventions applied consistently across all records - functioning as a content harmonisation instrument.	► Improves catalogue discoverability without post-hoc normalisation
Schema migration	INPUT Dataset + Supporting documents + Organisational attributes + Existing record (any schema)	HOW Legacy record parsed into field candidates and indexed in the retrieval corpus. Mapping from source to target schema fields emerges from the LLM's semantic knowledge - no explicit transformation rules required.	► Converts to target schema with no pre-defined mapping
Enrichment	INPUT Dataset + Supporting documents + Organisational attributes + Partial record	HOW Declared organisational attributes (publisher, license, contact) merged at extraction time and excluded from LLM generation. Remaining free-text fields generated and harmonised through the prompt template.	► Useful for metadata updates and profile completions

Companion Study: Prompt Engineering

Towards AI-Assisted Metadata Generation For Improved Description of Geospatial Data - submitted to Open Research Europe

Research questions:

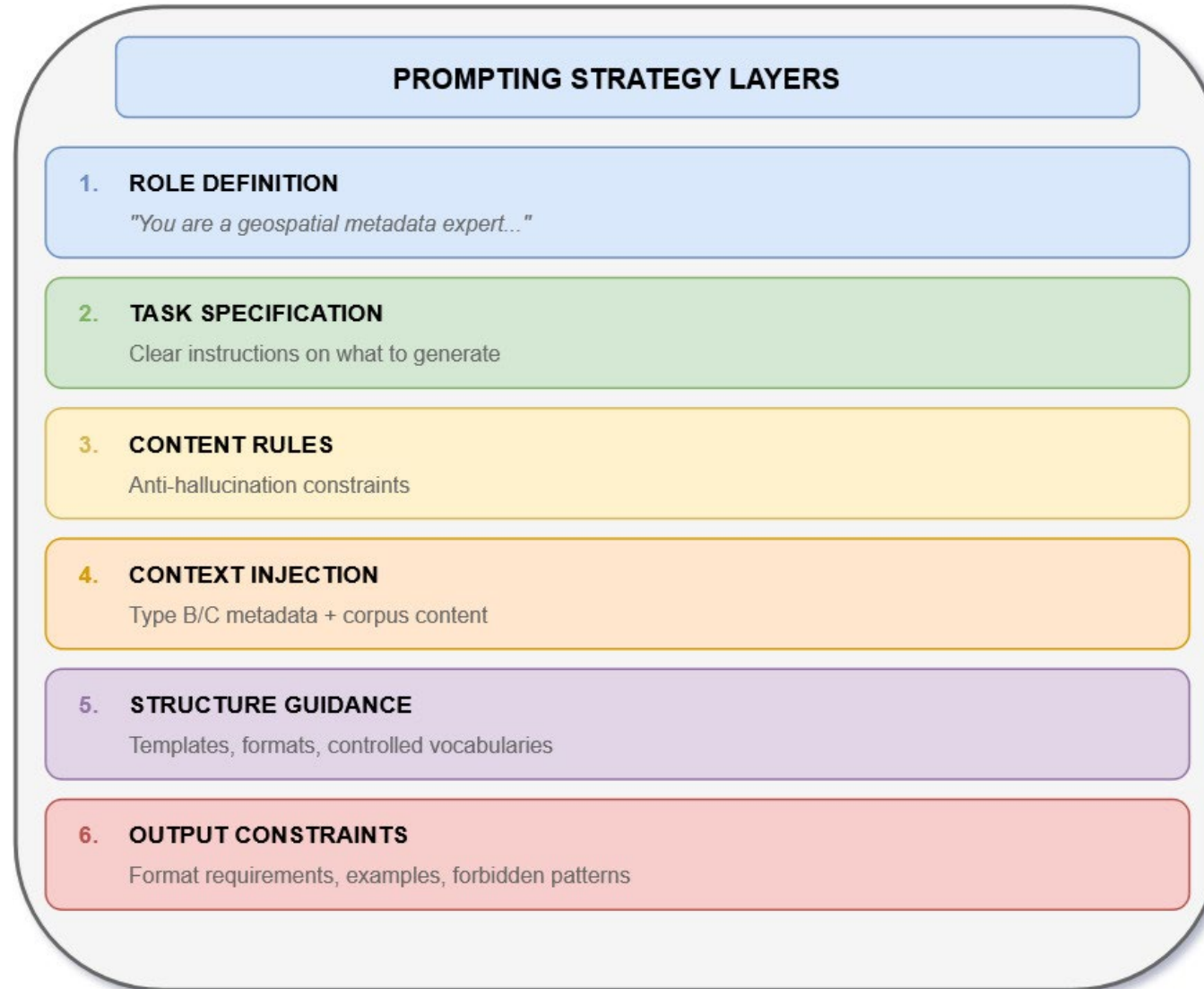
- To what extent can layered prompt engineering improve the **quality** of LLM-generated geospatial metadata **description field**?
- How do individual prompt components interact with model selection to affect structural compliance, semantic similarity and factual reliability?

Method

- 6-layer prompting framework · 7 ablation strategies · 2 LLMs (qwen3-32b and qwen3-coder-30b-a3b-instruct) · 8 datasets -- 112 combinations in total
- Evaluated using 10 automated metrics (lexical · structural · semantic) + human expert factual accuracy assessment

Six-layer prompting framework

Prompting Strategy Layers



Seven ablation strategies

Prompting Strategy Layer Configuration Matrix

Green = Enabled | Red = Disabled | Blue = Fixed (same across all strategies)

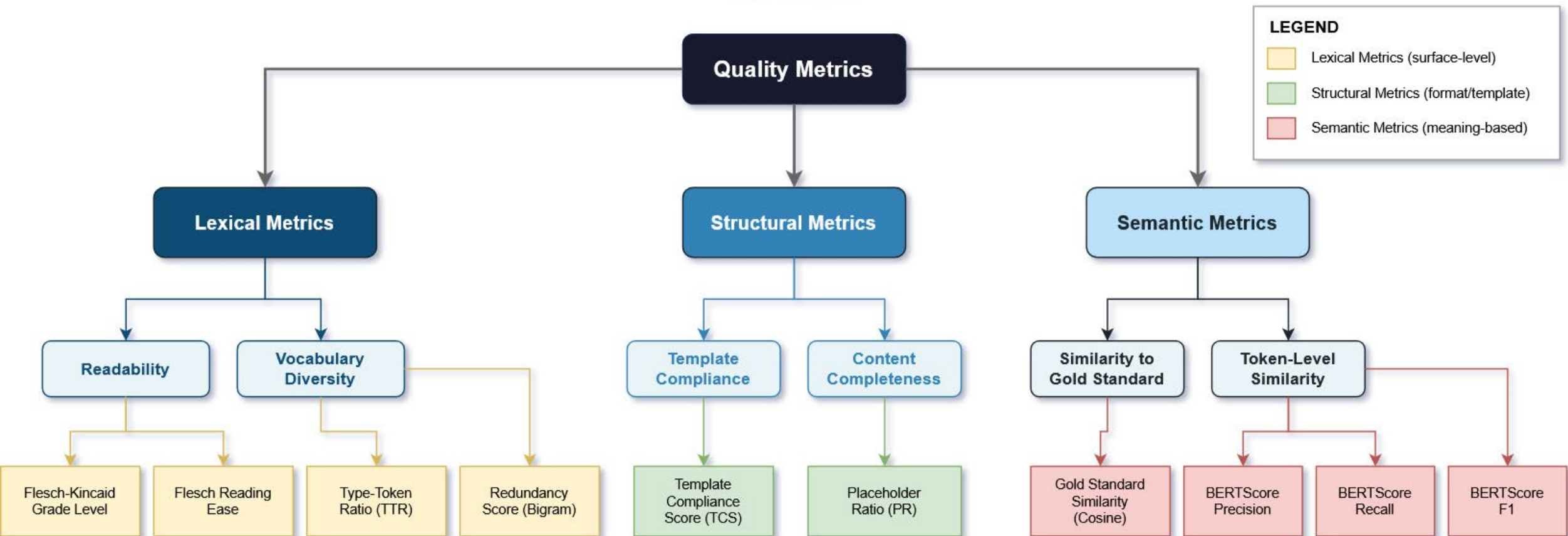
LAYERS	S1 Full Expert (Reference)	S2 No Role (Role Test)	S3 Expert Minimal (Examples Test)	S4 Expert No-Rules (Rules Test)	S5 Expert Free-Form (Template Test)	S6 Expert Pos-Only (Neg Ex Test)	S7 Baseline (Lower Bound)
1. Role Definition Expert / Machine / None	EXPERT	NONE	EXPERT	EXPERT	EXPERT	EXPERT	NONE
2. Task Specification FIXED - Same for all	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED
3. Content Rules Anti-hallucination constraints	✓ ON	✓ ON	✓ ON	X OFF	✓ ON	✓ ON	X OFF
4. Context Injection FIXED - Type B/C + Corpus	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED	✓ FIXED
5. Structure Guidance Template + Extraction Rules	FULL Template + Rules	FULL Template + Rules	PARTIAL Template only	FULL Template + Rules	X OFF Free-form	FULL Template + Rules	X OFF Free-form
6. Output Constraints Positive + Negative Examples	FULL Pos ✓ + Neg ✓	FULL Pos ✓ + Neg ✓	X OFF Pos X + Neg X	FULL Pos ✓ + Neg ✓	X OFF Pos X + Neg X	PARTIAL Pos ✓ + Neg X	X OFF Pos X + Neg X

Legend: Green Enabled / Full Yellow Partial / Changed Red Disabled / Off Blue Fixed (constant across all)

Key Comparisons: S1 vs S2 (Role) | S1 vs S3 (Examples) | S1 vs S4 (Anti-hallucination) | S1 vs S5 (Template) | S1 vs S6 (Neg Examples) | S1 vs S7 (Total Impact)

Evaluation metrics

METADATA QUALITY METRICS TAXONOMY



Key Findings

Lexical Metrics

	FK Grade	Reading Ease	TTR	Redundancy
S1 (32b)	12.9	26.1	0.673	0.078
S1 (coder-30b)	12.2	29.6	0.694	0.061
S2 (32b)	12.0	30.4	0.692	0.080
S2 (coder-30b)	12.4	28.5	0.699	0.056
S3 (32b)	13.9	23.0	0.547	0.163
S3 (coder-30b)	12.3	32.0	0.433	0.353
S4 (32b)	12.2	28.5	0.702	0.062
S4 (coder-30b)	12.6	27.4	0.694	0.061
S5 (32b)	15.5	12.0	0.641	0.088
S5 (coder-30b)	15.1	20.2	0.663	0.061
S6 (32b)	11.9	31.0	0.697	0.079
S6 (coder-30b)	12.0	30.7	0.712	0.055
S7 (32b)	16.7	2.2	0.635	0.124
S7 (coder-30b)	15.2	17.2	0.578	0.160



Structural Metrics

	Template Compliance	Placeholder Ratio
S1 (32b)	0.975	0.100
S1 (coder-30b)	0.992	0.008
S2 (32b)	0.975	0.108
S2 (coder-30b)	0.975	0.000
S3 (32b)	0.708	0.108
S3 (coder-30b)	0.742	0.150
S4 (32b)	0.983	0.050
S4 (coder-30b)	0.983	0.017
S5 (32b)	0.850	0.008
S5 (coder-30b)	0.792	0.017
S6 (32b)	0.950	0.092
S6 (coder-30b)	0.992	0.008
S7 (32b)	0.733	0.000
S7 (coder-30b)	0.733	0.142



Semantic Metrics

	Gold Std Similarity	BERTScore Precision	BERTScore Recall	BERTScore F1
S1 (32b)	0.727	0.825	0.844	0.834
S1 (coder-30b)	0.733	0.835	0.841	0.837
S2 (32b)	0.712	0.827	0.835	0.830
S2 (coder-30b)	0.726	0.830	0.842	0.835
S3 (32b)	0.719	0.792	0.841	0.815
S3 (coder-30b)	0.710	0.797	0.840	0.817
S4 (32b)	0.700	0.832	0.838	0.835
S4 (coder-30b)	0.729	0.832	0.840	0.836
S5 (32b)	0.744	0.804	0.854	0.828
S5 (coder-30b)	0.721	0.819	0.851	0.834
S6 (32b)	0.732	0.827	0.840	0.833
S6 (coder-30b)	0.728	0.829	0.840	0.833
S7 (32b)	0.749	0.787	0.847	0.816
S7 (coder-30b)	0.753	0.805	0.845	0.824



Key Findings

Lexical metrics

The template doesn't only impose structure - it disciplines the prose.

With template (S1, S2, S4, S6) the descriptions read as clean technical text; without it (S5, S7) the models drift into convoluted sentences (Flesch-Kincaid 15-17, Reading Ease below 20) that are hard to parse even for experts.

Remove the template, and readability collapses; remove the rules and examples, and the model repeats itself

Structural metrics

Template Compliance behaves as expected: with a template, records conform almost perfectly (0.95-0.99); without one, they fall out of range (0.71-0.85).

But Placeholder Ratio is easy to misread. The coder model looks better - almost no gaps (≤ 0.02) versus the general model's visible ones (0.05-0.11).

Yet, fewer placeholders isn't more complete metadata: the coder hits those numbers by quietly dropping or inventing content, while the general model honestly marks what it couldn't find. The metric that looks worse is the safer behaviour.

Semantic metrics

BERTScore is uniformly strong (F1 0.82-0.84): the meaning is captured whatever the prompt.

Gold Standard Similarity rates the free-form strategies (S5, S7) highest because their phrasing sounds more "human", while the fully structured strategies (S1, S2, S4, S6) have highest BERTScore because it checks whether the same information is present.

Key takeaways

- ✓ We built an open-source, schema-agnostic metadata editor for geospatial data: it drafts standards-compliant records automatically, with a **human** reviewing before export.
- ✓ The **template** is the most important design choice. It makes the generated descriptions readable, well-structured, and consistent from one record to the next; examples and rules then keep the text concise and non-repetitive.
- ✓ Honesty about **missing information** matters more than apparent completeness. Explicit "[Not specified]" placeholders flag gaps rather than hiding or inventing content - so the more complete-looking output can be the one needing more checking. The general-purpose model (which uses placeholders) is preferable to the code-specialised one (which avoids them).
- ✓ Output quality is bounded by the quality of the input documentation: good source material in, good metadata out.

Next steps

- ✓ Planning an experimentation of metadata enrichment in collaboration with CNR and GEODAB
- ✓ Experimentation of schema remapping in collaboration with DG DIGIT - SEMIC
- ✓ Next developments will include a broader schema coverage
- ✓ Community validation within INSPIRE MIG-T Pilot

Thank you



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